

PREDICTING SCHOOL BUS USAGE

Who Rides the Bus in Abu Dhabi and Why It Matters

A data-driven view of 4000+ families across Abu Dhabi, Al Ain, and Al Dhafra

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Can we predict and explain bus usage?

Why this matters

Morning Traffic

Every family that drives to school is a car on the road in peak hours.

Funding allocation

Identify where buses are underused and could be improved?

Expansion

ADEK needs to know which communities to engage and how.

THE PROJECT

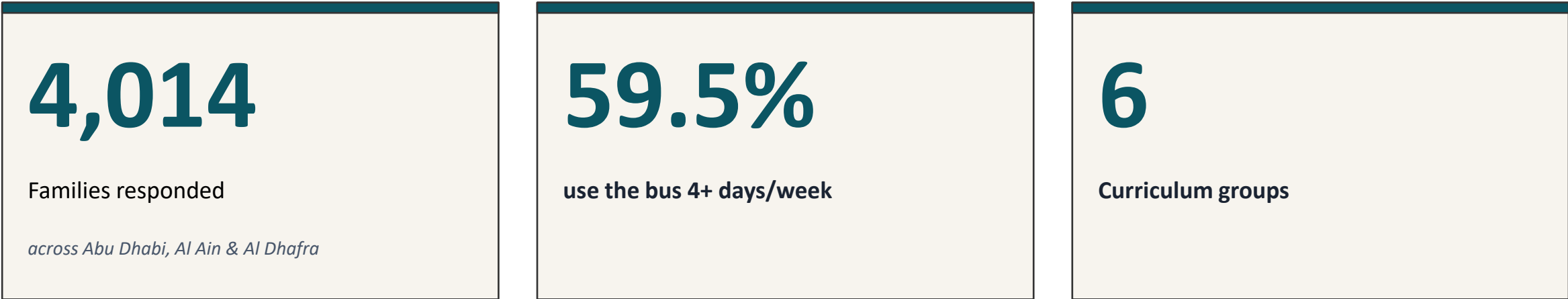
Predict the probability that a child uses the school bus 4+ days per week and explain what drives it.

Survey: 4000+ families and 96 variables

Coverage: Abu Dhabi, Al Ain, Al Dhafra

Outcome: Probability of using the bus services frequently

Survey responses indicate around 60% of the families use the bus frequently



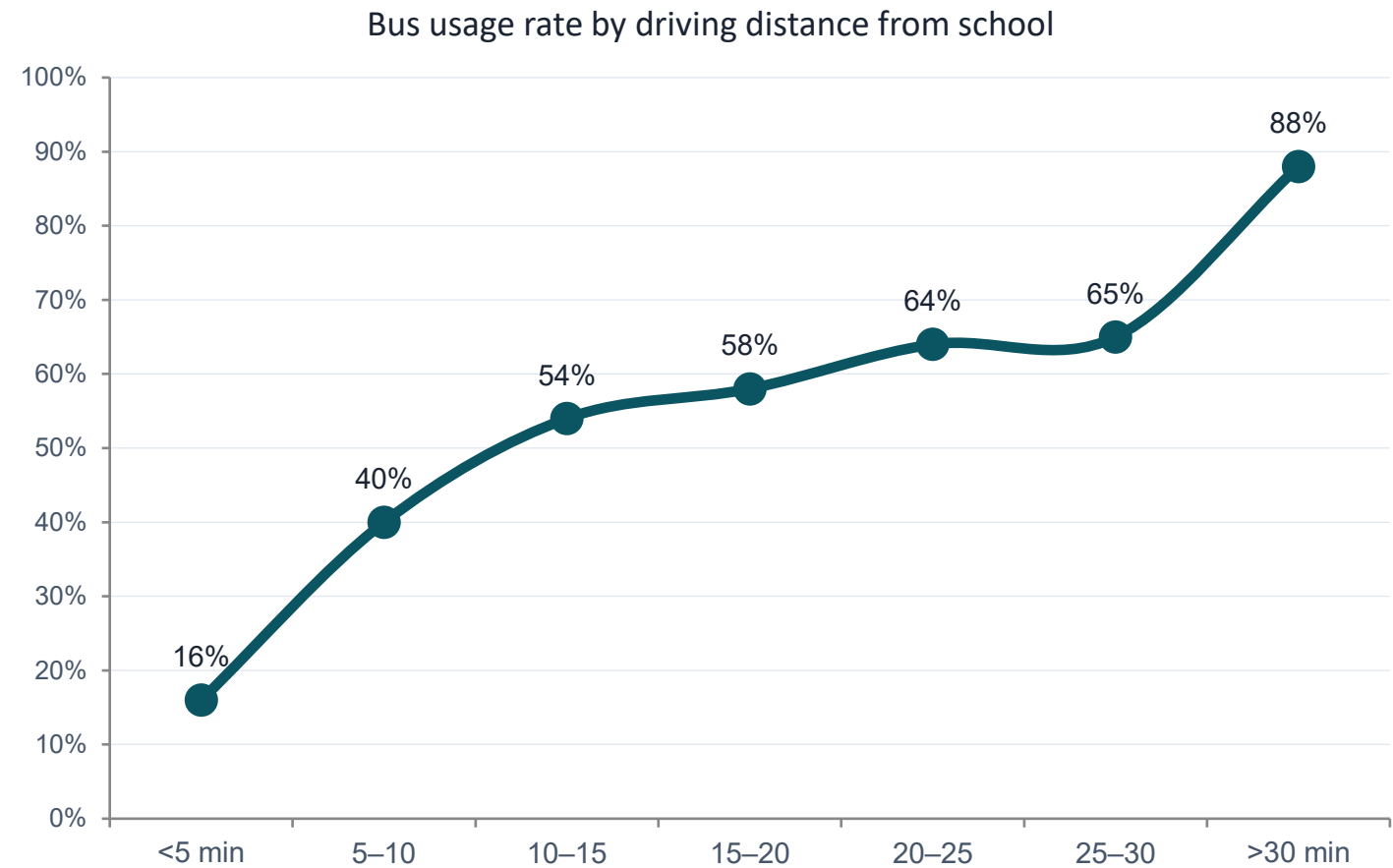
What predicts whether a family uses the bus?

Transportation: Distance to School, Bus satisfaction, Number of cars, Mother / Father license

Demographics: Income, Housing type, Neighborhood

School factors: Curriculum group

The most intuitive factor in predicting the bus usage



From 16% to 88%

Bus usage rises greatly with distance.

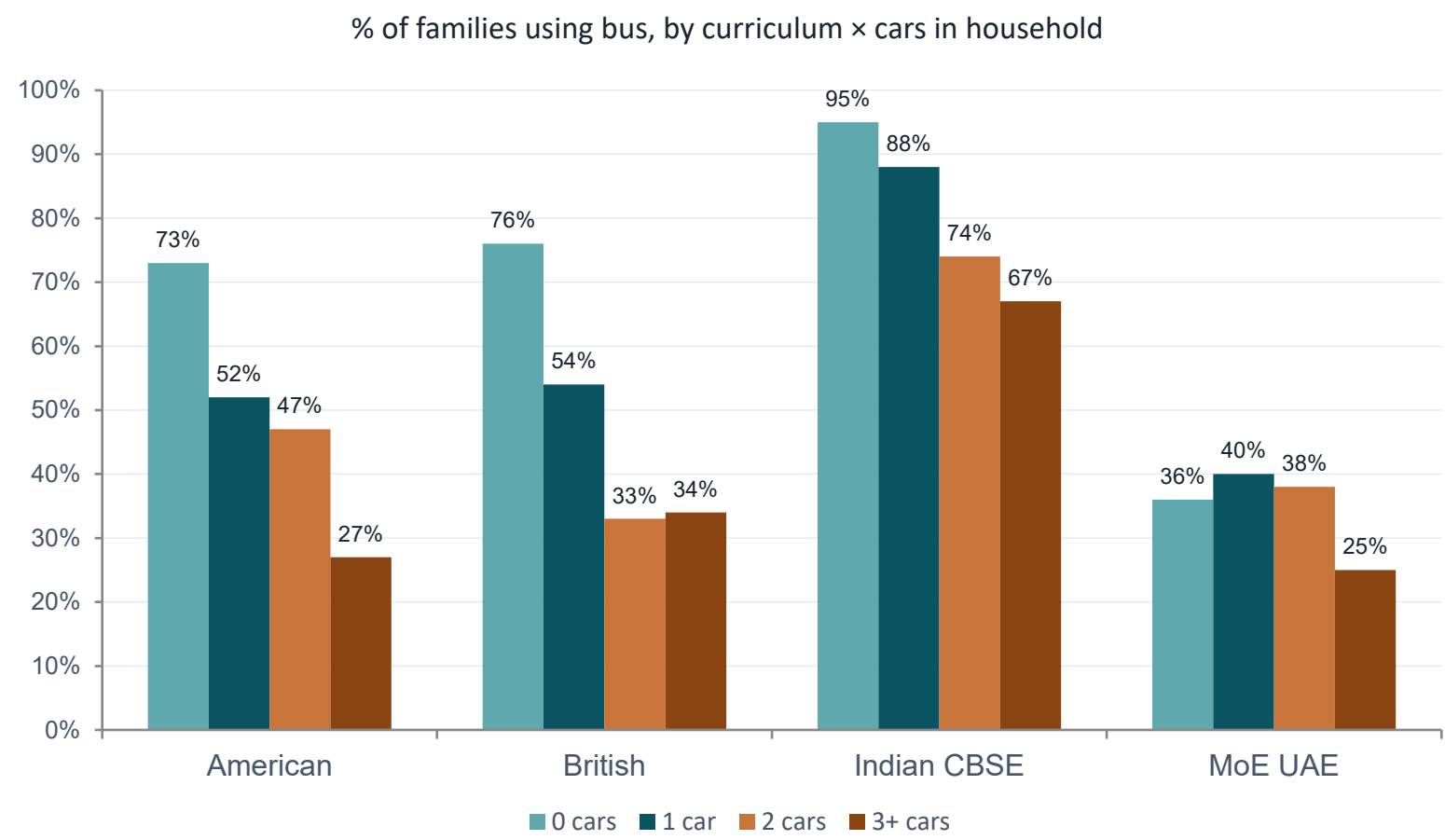
Families <5 min away:

Kids walk, take a bicycle or get dropped off

Families >30 min away:

**an hour-plus daily round-trip
the bus is the rational choice.**

Similar families behave differently by curriculum



Indian CBSE families with 3+ cars still use the bus 67% of the time.

Other curriculum families with 3+ cars: only around 30%.

Same logistics. Same wealth. Different choice.

→ Bus uptake isn't only about distance and cars
community norms and school-organized transport play a big role.

Bus usage prediction: Model comparison

Logistic Regression

Predicts the probability of bus usage based on given variables.

STRENGTH

Interpretable. Statistically significant predictors are easy to identify.

LIMITATION

Performs poorly with complex patterns or interactions between variables.

Random Forest

Combines 500 decision trees to make more accurate predictions.

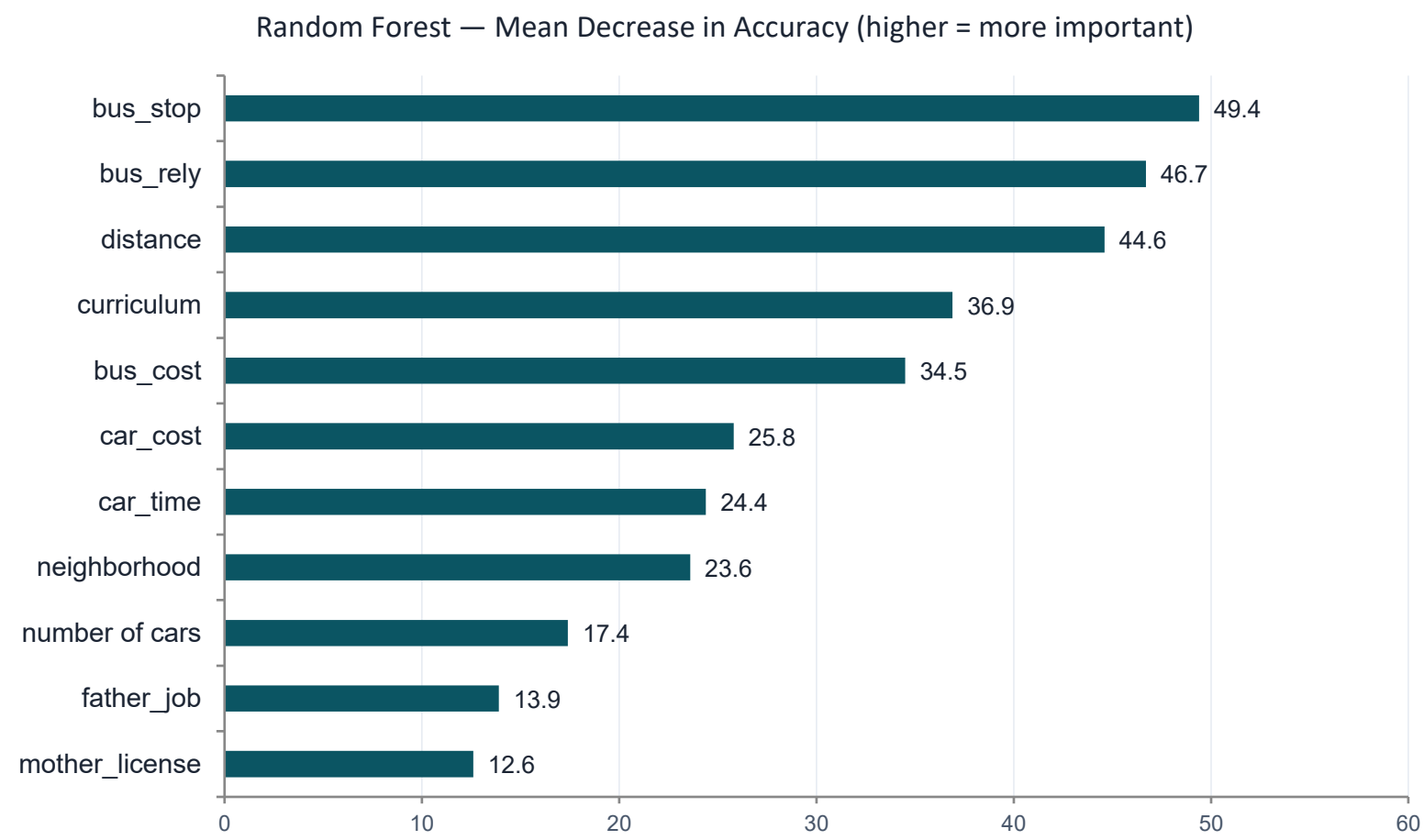
STRENGTH

Automatically captures complex patterns in the data.

LIMITATION

Less directly interpretable but provides variable importance scores.

The strongest predictors in our model



TOP 3 PREDICTORS

1

Bus_stop
Bus stop locational satisfaction

2

Bus_rely
Bus reliability satisfaction

3

Distance
Distance from home to school

What the models actually tell us

45×

more likely to use the bus

A family living >30 min away has around 45× higher odds of frequent bus use than a family <5 min away

Indian CBSE

highest curriculum effect

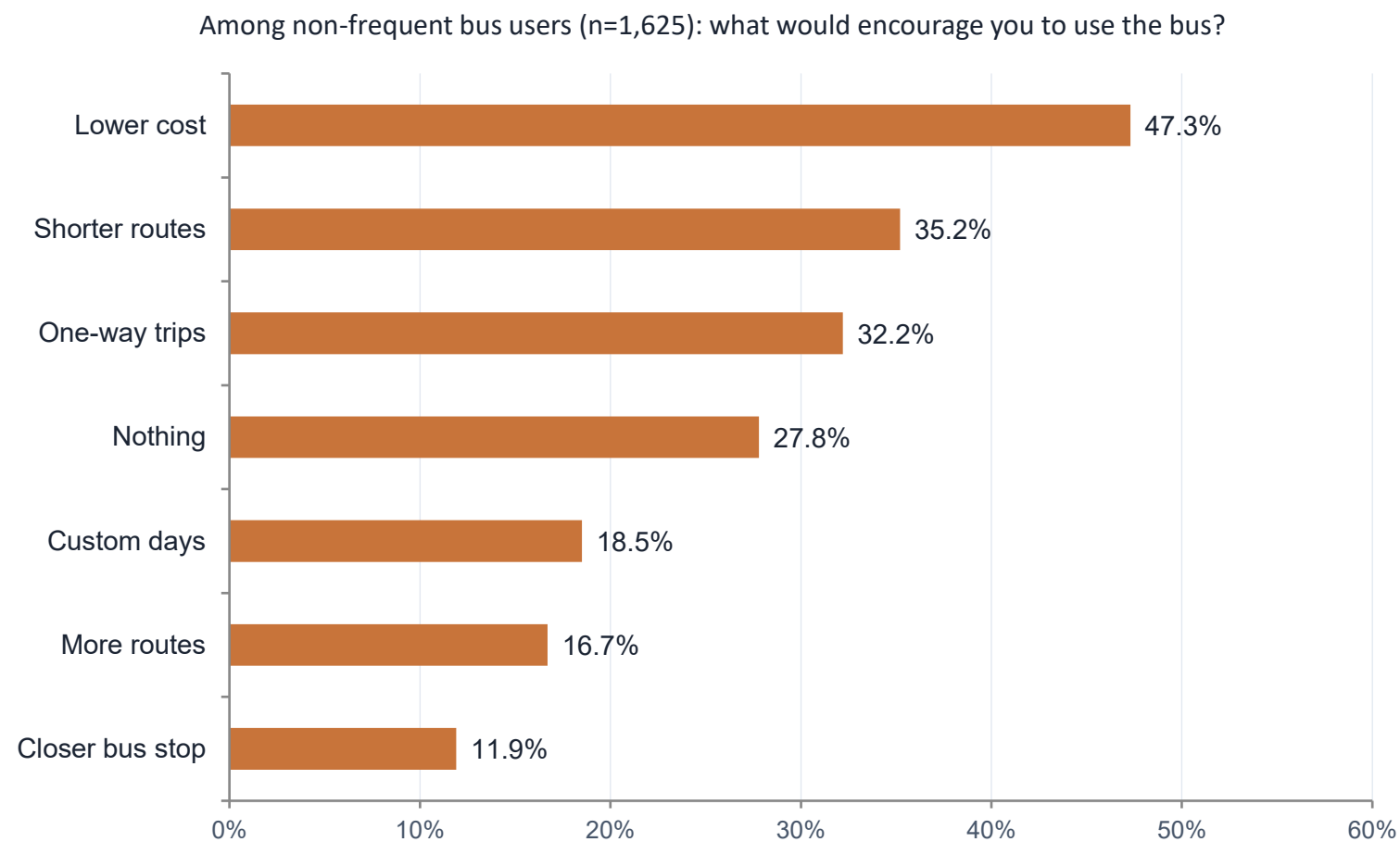
Holding distance, cars, and income constant, Indian CBSE families have multiple-times higher odds of frequent bus use than American curriculum families.

85%

Random Forest accuracy

On a held-out test set (20% of families never seen during training), the Random Forest correctly classifies around 85% of families meanwhile Logistic Regression classifies around 73%.

Non-bus using families often identify the same limitations



KEY SIGNALS

Cost + flexibility

47%
want lower cost

32%
want one-way options

19%
want custom days

Three actions for ADEK + ITC

1

Include flexible, partial-week bus subscriptions

Test one-way trips, custom-day plans, and proportional pricing — directly responding to the top survey signals

2

Engage non-Indian curriculum communities

Indian CBSE bus uptake is 4× higher even controlling for distance and cars. Study what makes the bus work so well for Indian CBSE families and replicate.

3

Shorten the routes

35% of non-bus using families indicate shorter routes would encourage them to use the bus more frequently.

What we can and can't say from this analysis

Curriculum imbalance

Indian CBSE and British families make up about 70% of respondents. Conclusions about smaller groups come from fewer respondents and should be treated as suggestive.

Self-reported responses

Families chose to respond. Highly-engaged or strongly-opinionated families may be over-represented.

Categorical distance

Distance is approximate, we have driving time in 5-minute intervals, whereas distance in km would let us see the threshold more precisely.

Predictive, not causal

Both models predict who uses the bus. Neither proves that, for example, changing curriculum would change bus use.

No single factor explains bus usage, distance curriculum and community all play a role.

The data points to real opportunities and families have told us exactly what would help.

Include flexible subscriptions

Engage underused curriculum groups

Shorten the routes

Thank you.

Questions?

Confusion matrix and test-set metrics

Logistic Regression

Confusion matrix (test set, n=803):

	Pred: no	Pred: yes
Actual: no	215	110
Actual: yes	109	369

Accuracy

72.7%

Sensitivity

77.2%

Specificity

66.2%

Precision

77.0%

Kappa

0.434

Random Forest

Confusion matrix (test set, n=803):

	Pred: no	Pred: yes
Actual: no	250	75
Actual: yes	47	431

Accuracy

84.8%

Sensitivity

90.2%

Specificity

76.9%

Precision

85.2%

Kappa

0.680

Methodology — what we improved

Imputation, not row-dropping

Before: `na.omit()` dropped around 10% of rows (437 of 4,014).

After: median imputation for numeric vars; mode imputation for categorical vars (incl. `father_job`, `mother_job`). All 4,014 rows retained.

Categorical encoding for the Random Forest

Before: `father_job` and `mother_job` entered the forest as numeric codes.

After: both factored before fitting. RF now receives categorical variables, as intended.

80/20 stratified train/test split

Before: both models evaluated on training data which is a major issue, as it might perform well on the training data, and poorly on an unseen one.

After: Used train/test split so the models learn patterns and output real-life accuracy, rather than memorizing the data.